

There are 10 soldiers in a small barracks. For each soldier there is a probability of 0.02 (2%) of contracting a variant of Covid-19 (independently from each other). The doctor wants to determine whether any of them have that Covid-19 variant. Since the tests are hard to come by, the doctor mixes together blood samples from all the 10 soldiers and performs a test on the mixed blood. If any one of the soldiers has Covid-19 variant the mixed blood will test positive.

(a) What is the probability the mixed blood will test positive?

let $P(s)$ represent the probability that a soldier tests positive let $P(a)$ represent the probability that the mixed solution tests positive

$$P(a) = 1 - P(s^c)^{10} = 1 - (1-0.02)^{10} = 1 - 0.182927193$$

(b) John is one of the soldier among the ten. What is the probability that John has the disease if the mixed blood test is positive?

let $P(j)$ represent the probability that john tests positive

$$P(j | a) = P(a | j) * P(j) / P(a) \quad P(a | j) = 1 \quad P(j) = 0.02 \quad P(a) = 0.182927193$$

$$P(j | a) = 0.109333116$$

(c) If the mixed blood is positive, the doctor will test individually all the ten soldiers. What is the expected total number of Covid-19 tests that the doctor has to perform?

$$E(\text{tests}) = 1 * 0.8171 + 11 * 0.1829$$

$$E(\text{tests}) = 0.8171 + 2.0119 = 2.829$$

So the doctor is expected to perform about **2.83 tests**.

A call center company is interested in modeling the number of calls that they receive in a 10 minute interval in order to determine how many operators to hire. The list[int] `call_center_data` contains the some data they collected in the month of October on the number of calls received in 10 minute intervals.

```
In [1]: call_center_data = [4,9,8,10,11,11,14,10,5,15,10,7,7,10,9,6,6,4,5,10,7,3,2,1
```

(a) Let X be the number of calls that arrive at the call center in a 10 minute interval. Find the observed probability (relative frequency) distribution of X from the data collected in October.

```
In [ ]: from collections import Counter

observed_distribution = {
    k: v / len(call_center_data) for k, v in sorted(Counter(call_center_data
```

```
}  
observed_distribution
```

```
Out[ ]: {0: 0.004629629629629629,  
1: 0.02546296296296296,  
2: 0.04398148148148148,  
3: 0.05092592592592592,  
4: 0.08101851851851852,  
5: 0.08333333333333333,  
6: 0.08333333333333333,  
7: 0.1087962962962963,  
8: 0.08564814814814815,  
9: 0.10416666666666667,  
10: 0.09027777777777778,  
11: 0.06712962962962964,  
12: 0.04861111111111111,  
13: 0.037037037037037035,  
14: 0.03009259259259259,  
15: 0.023148148148148147,  
16: 0.013888888888888888,  
17: 0.009259259259259259,  
18: 0.004629629629629629,  
19: 0.0023148148148148147,  
21: 0.0023148148148148147}
```

(b) Calculate the average number λ of calls received per 10 minute interval in October

```
In [3]: from statistics import mean  
  
mean(call_center_data)
```

```
Out[3]: 7.921296296296297
```

(c) Randomly occurring telephone calls are often modeled by a Poisson distribution. Calculate the theoretical probability distribution of X assuming a Poisson distribution with mean λ that you calculated in part (b).

```
In [ ]: from scipy.stats import poisson  
  
poisson_distribution = {  
    k: poisson.pmf(k, mu=mean(call_center_data)) for k in observed_distribut  
}  
poisson_distribution
```

```
Out[ ]: {0: np.float64(0.0003629315546010011),
1: np.float64(0.0028748883792699687),
2: np.float64(0.011386421335488226),
3: np.float64(0.0300650723842907),
4: np.float64(0.059538586631390465),
5: np.float64(0.09432455715398985),
6: np.float64(0.12452879420561482),
7: np.float64(0.14091849661759712),
8: np.float64(0.13953214566707667),
9: np.float64(0.12280838540965473),
10: np.float64(0.0972801608499623),
11: np.float64(0.0700531798039923),
12: np.float64(0.04624266614376205),
13: np.float64(0.028177066158111375),
14: np.float64(0.015942777842767344),
15: np.float64(0.008419164471905818),
16: np.float64(0.004168168521826062),
17: np.float64(0.0019421939926046862),
18: np.float64(0.0008547052266838046),
19: np.float64(0.00035633543929239255),
21: np.float64(5.3235611077057785e-05)}
```

(d) Plot the two probability distributions (a) and (c) together on the same plot. Looking at the plot, decide whether our assumption that X follows a Poisson distribution is reasonable or not.

```
In [ ]: import matplotlib.pyplot as plt

# Convert to lists for plotting
observed_probs = [observed_distribution[i] for i in observed_distribution]
poisson_probs = [poisson_distribution[i] for i in observed_distribution]

plt.figure(figsize=(10, 6))
plt.bar(
    observed_distribution.keys(),
    observed_distribution.values(),
    label="Observed",
    align="center",
)
plt.plot(
    observed_distribution.keys(),
    poisson_distribution.values(),
    label="Poisson",
    color="red",
)
plt.xlabel("Calls")
plt.ylabel("Probability")
plt.title("Observed vs Theoretical Distribution")
plt.legend()
plt.show()
```

